Plugging into the Future: An Exploration of Electricity Consumption Patterns

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1. INTRODUCTION

1.1 Project Overview

Electricity plays a crucial role in economic growth and development. The consumption of electricity reflects industrial activity, population demand, and infrastructure development. This project focuses on analyzing electricity consumption patterns across Indian states over a period of 24 months, from January 2019 to December 2020.

The dataset consists of 16,600 records, covering 33 states and 5 regions of India. It includes attributes such as State, Region, Latitude, Longitude, Date, and Usage (in Mega Units).

The project uses **Tableau** for creating interactive dashboards, **MySQL** for basic data handling and validation, and **GitHub** for version control and project demonstration. The primary focus is to identify trends, regional differences, and the impact of the COVID-19 lockdown on electricity consumption.

1.2 Purpose

The main objectives of this project are:

- To analyze electricity consumption trends over time
- To compare electricity usage across different regions and states
- To study the impact of COVID-19 lockdown on electricity consumption
- To identify top and bottom performing states
- To create interactive dashboards for better visualization and decision-making

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2. IDEATION PHASE

2.1 Problem Statement

Electricity consumption data is large, complex, and difficult to interpret using traditional tools like Excel. Analysts face challenges in comparing data across states and identifying trends over time. Additionally, there is no interactive system available to analyze the impact of COVID-19 on electricity consumption.

2.2 Empathy Map Canvas

The primary users of this system are energy analysts, government officials, and data analysts.

- **Think & Feel:** Need accurate and fast insights; concerned about data complexity
- **See:** Raw datasets, reports, and static charts
- **Hear:** Demand for faster reporting and better tools
- **Say & Do:** Analyze data manually, create reports
- **Pain:** Time-consuming analysis, lack of visualization
- **Gain:** Interactive dashboards, quick insights, better decision-making

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2.3 Brainstorming

Various ideas were generated to solve the problem:

- Time-series analysis of electricity consumption
- Region-wise comparison
- State-wise consumption analysis
- COVID-19 lockdown impact analysis
- Top and bottom state identification
- Dashboard creation with filters

These ideas were grouped into categories such as trend analysis, comparison analysis, impact analysis, and visualization. Based on feasibility and impact, the most effective ideas were selected for implementation.

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3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey represents how users interact with the system:

- **Entice:** User identifies need for electricity analysis
- **Enter:** User opens the Tableau dashboard
- **Engage:** User applies filters and analyzes data
- **Exit:** User interprets results and insights
- **Extend:** User revisits dashboard for further analysis

3.2 Solution Requirements

Functional Requirements:

- Load and process dataset
- Create visualizations
- Provide filters (Year, Region, State)
- Analyze trends and comparisons
- Display COVID-19 impact

Non-Functional Requirements:

- Fast performance
- User-friendly interface
- Accurate results
- Scalability for future data

3.3 Data Flow Diagram

The data flow of the system is as follows:

CSV Dataset → MySQL Database → Tableau Processing → Dashboard → User

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3.4 Technology Stack

- **Tableau:** Data visualization and dashboard creation
- **MySQL:** Data storage and querying
- **Python:** Data validation (optional)
- **GitHub:** Version control and project management

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4. PROJECT DESIGN

4.1 Problem-Solution Fit

The problem of analyzing large and complex electricity data is solved by transforming raw data into interactive visual dashboards. This allows users to easily understand trends and make data-driven decisions.

4.2 Proposed Solution

The proposed solution includes:

- Importing dataset into MySQL
- Connecting Tableau with data source
- Creating calculated fields (Year, Month, Quarter, Lockdown Phase)
- Building multiple visualizations
- Designing interactive dashboards
- Creating a story for step-by-step insights

4.3 Solution Architecture

The system follows a three-layer architecture:

1. **Data Layer:** CSV file and MySQL database
2. **Processing Layer:** Tableau calculations and analysis
3. **Presentation Layer:** Dashboards and story

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5. PROJECT PLANNING & SCHEDULING

The project was developed using an agile approach with multiple phases:

- Data collection and cleaning
- Data preprocessing and calculations
- Visualization creation
- Dashboard design
- Story creation
- Testing and validation

Each phase was completed in a structured manner to ensure timely delivery.

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6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

- Total dataset size: 16,600 rows
- Number of visualizations: 12
- Number of dashboards: 3
- Number of story scenes: 15
- Filters used: Year, Region, State, Lockdown Phase

The system was tested for performance, accuracy, and usability. All visualizations and filters worked correctly without errors.

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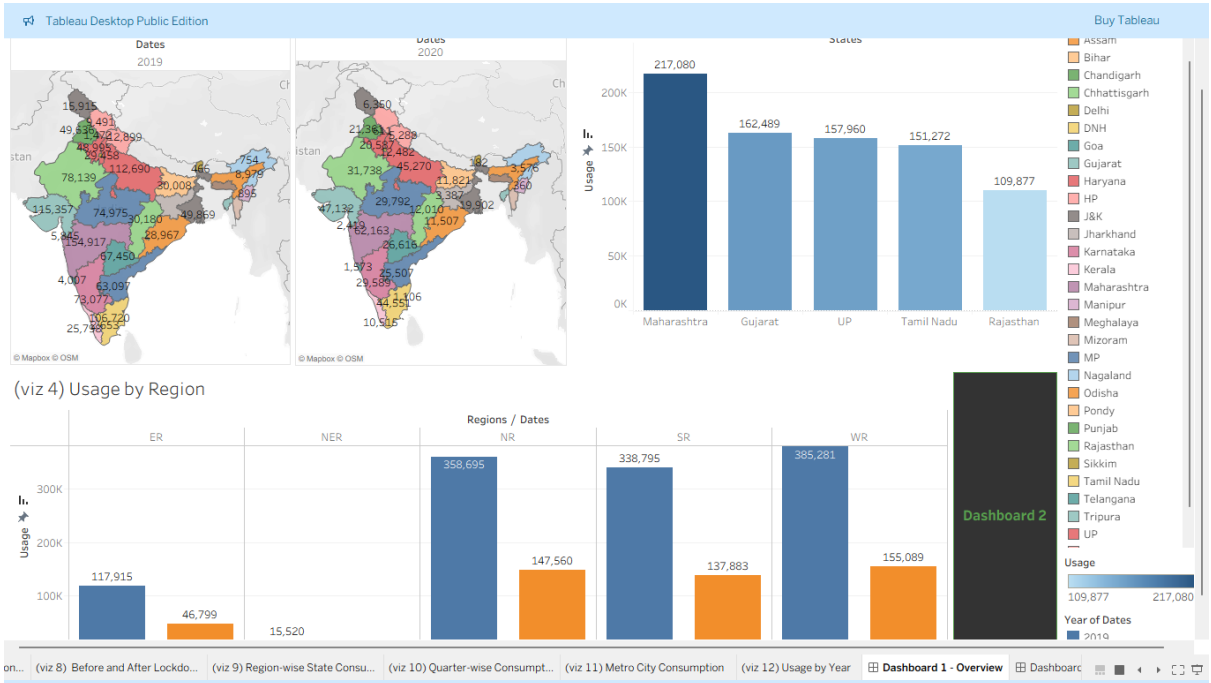
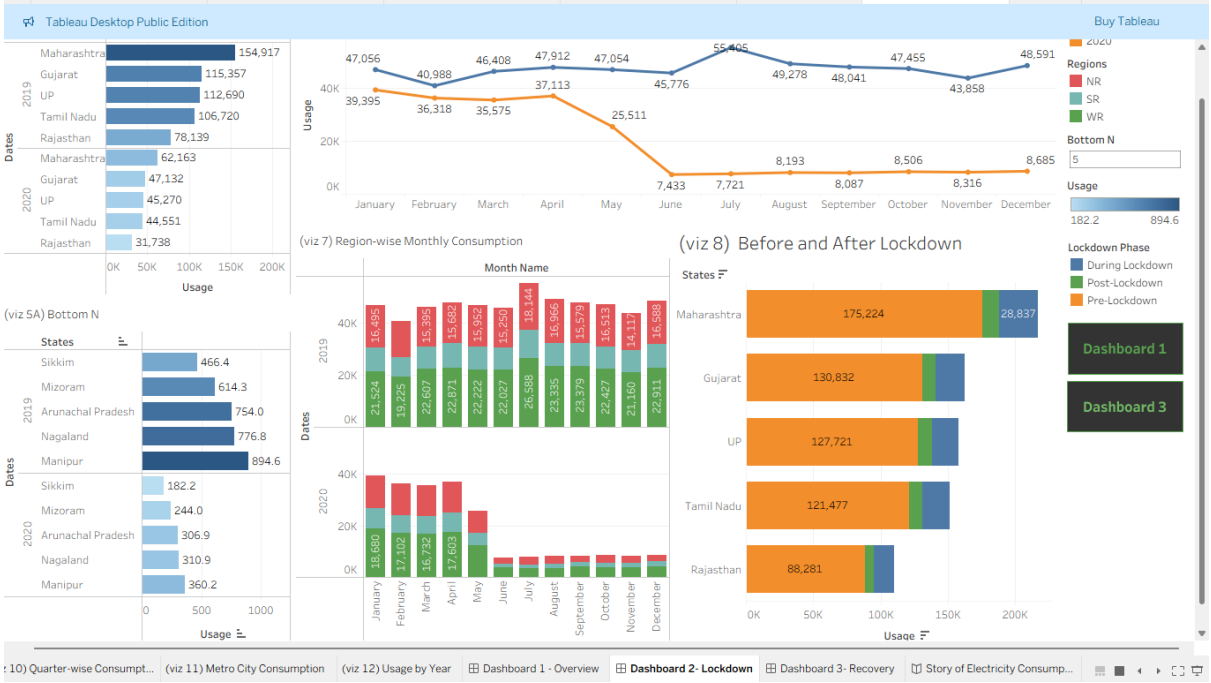
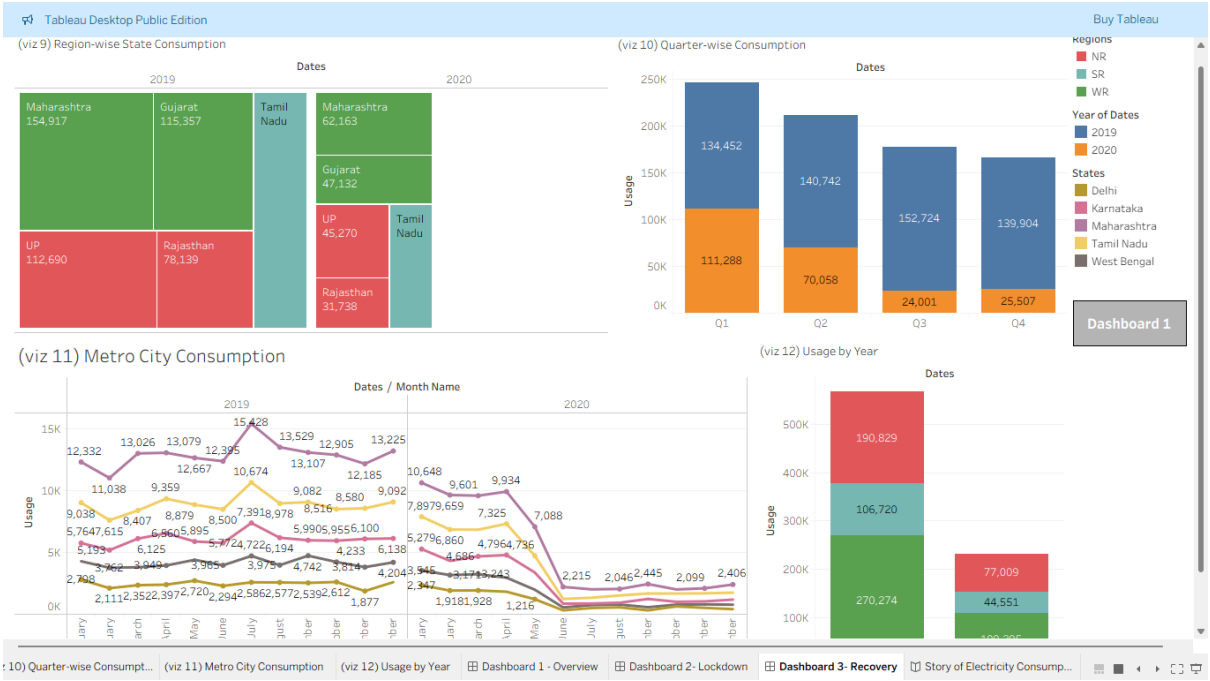
7. RESULTS

7.1 Output Screenshots







Key Findings:

- Maharashtra has the highest electricity consumption
- Significant drop observed during COVID-19 lockdown (March–June 2020)
- Western region shows highest usage
- Gradual recovery observed after lockdown
- Seasonal trends visible in summer months

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8. ADVANTAGES & DISADVANTAGES

Advantages:

- Interactive and easy-to-use dashboards
- Quick analysis and insights
- Better decision-making
- Visual representation of data

Disadvantages:

- Requires knowledge of Tableau
- Dependent on dataset quality
- Limited predictive capability

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9. CONCLUSION

This project successfully analyzes electricity consumption patterns across India and highlights the impact of COVID-19 lockdown. The use of Tableau dashboards provides clear and interactive insights, making complex data easier to understand and analyze.

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10. FUTURE SCOPE

- Integration of real-time data
- Implementation of predictive analytics
- Enhancement of dashboard design
- Mobile-friendly dashboard development

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11. APPENDIX

- Dataset Source: POSOCO

- Tableau Public Link: https://public.tableau.com/app/profile/faisal.khan7191/viz/ElectricityConsumptionAnalysis-Story/StoryofElectricityConsumptioninIndia?publish=yes

- GitHub Repository: https://github.com/faisalkhan02/Electricity-Consumption-India-Analysis-Smartbridge-Project